REFONTE INTERNATIONAL TRAINING AND INTERNSHIP PROGRAM

Astrodynamics Specialist

SESSION 8

GNSS Constellation Dynamics and Navigation

How GPS, Galileo and Their Peers Turn Orbits into Position Fixes

Prepared for Cao Khanh Tan

Module 8, in the Advanced Application Phase

Builds on Session 5 (constellations), Sessions 1 and 2 (orbits, perturbations)

Tools STK (GPS constellation, DOP and coverage), GMAT, Python

SESSION 8

GNSS Constellation Dynamics and Navigation

A satellite navigation system is a constellation with a job: to let a receiver anywhere on Earth work out where it is and what time it is. This module takes the constellation geometry of Session 5 and adds the navigation layer, how a receiver turns signals from several satellites into a position fix, why the geometry of those satellites decides the accuracy, and the orbital and relativistic effects that a navigation constellation must model to the metre. GPS, Galileo, GLONASS and BeiDou are the working examples throughout.

Learning objectives

- Describe the main GNSS constellations and explain why they sit in MEO.
- Explain the pseudorange positioning principle and why four satellites are needed.
- Define dilution of precision and relate constellation geometry to position accuracy.
- Identify the orbital perturbations and error sources a GNSS system must model.
- Compute the relativistic clock correction and its effect if ignored.
- Simulate satellite visibility and PDOP for a real constellation.

> **Where this connects**
> Session 5 asked how many satellites it takes to see a point. GNSS asks a sharper question: given the satellites in view, how well can a receiver be located? The answer depends on the same orbital dynamics you already know, now pushed to navigation-grade precision.

PART 1

The GNSS Constellations

The four global systems are all Walker constellations in medium Earth orbit, sized so that at least four satellites are visible from anywhere at any time:

System	Altitude / SMA	Inclination	Satellites	Period
GPS (USA)	~20,180 km / 26,560 km	55 deg	24+ (Walker 24/6)	11.97 h
GLONASS (Russia)	~19,130 km / 25,510 km	64.8 deg	24 (3 planes)	11.26 h
Galileo (EU)	~23,220 km / 29,600 km	56 deg	24 (Walker 24/3/1)	14.08 h
BeiDou (China)	MEO + IGSO + GEO mix	55 deg (MEO)	~30 (plus GEO/IGSO)	12.9 h (MEO)

Why MEO? It is the sweet spot for navigation. From ~20,000 km each satellite sees roughly a third of the Earth, so a few dozen satellites guarantee four or more in view everywhere, while the orbit is high enough to be stable and predictable but low enough for strong signals. GPS sits at almost exactly half a sidereal day, so it makes two revolutions per day and its ground track repeats daily, which keeps its coverage geometry predictable.

PART 2

How a Fix Is Computed

A receiver does not measure true range; it measures the signal travel time against its own imperfect clock, giving a pseudorange, the true range plus an unknown clock offset shared across all satellites:

$$\rho_i = \left| \mathbf{r}^i - \mathbf{r}_u \right| + c\,\delta t_u + \varepsilon_i$$

There are four unknowns, three position coordinates and the receiver clock bias, so a receiver needs signals from at least four satellites to solve for all of them:

$$4 \text{ unknowns}: \ (x,\, y,\, z,\, \delta t_u) \ \Rightarrow \ \text{need} \ \geq 4 \text{ satellites}$$

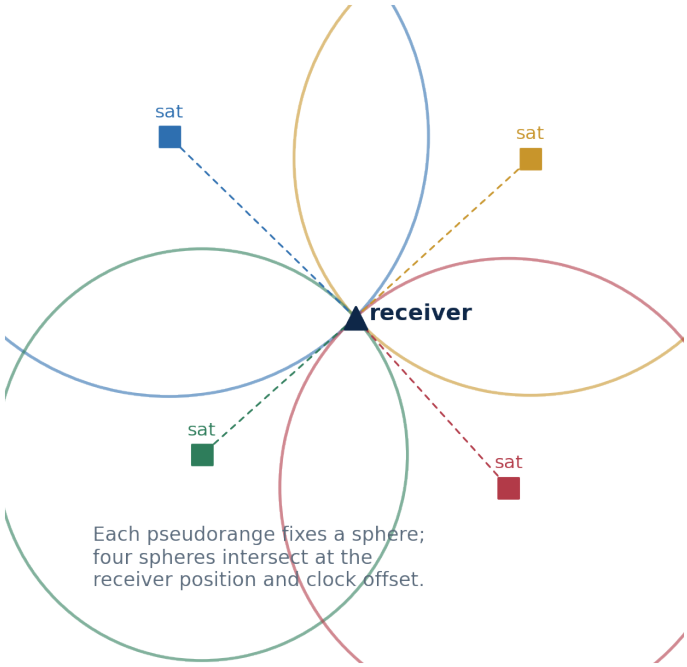


Figure 1. Positioning by pseudoranges. Each satellite fixes a sphere of possible positions; four spheres intersect at a single point and simultaneously solve the receiver clock error. This is why a GNSS receiver is also an excellent clock.

PART 3

Geometry Sets the Accuracy: Dilution of Precision

Having four satellites is necessary but not sufficient: their arrangement in the sky decides how much the ranging errors amplify into position error. Satellites spread widely across the sky give a sharp fix; satellites clustered together give a poor one. This amplification is the dilution of precision (DOP):

$$\sigma_{\text{pos}} = \text{PDOP} \times \sigma_{\text{UERE}}$$

DOP is purely geometric, computed from the line-of-sight directions to the visible satellites through the geometry matrix:

$$\text{PDOP} = \sqrt{Q_{xx} + Q_{yy} + Q_{zz}}, \quad Q = (H^T H)^{-1}$$

Variants isolate different parts: PDOP for 3D position, HDOP and VDOP for horizontal and vertical, TDOP for time, and GDOP for everything together. Lower is better; a PDOP under about 2 is excellent, above 6 is poor. Figure 2 shows both the number of satellites in view

and the PDOP for a GPS receiver over a full day.

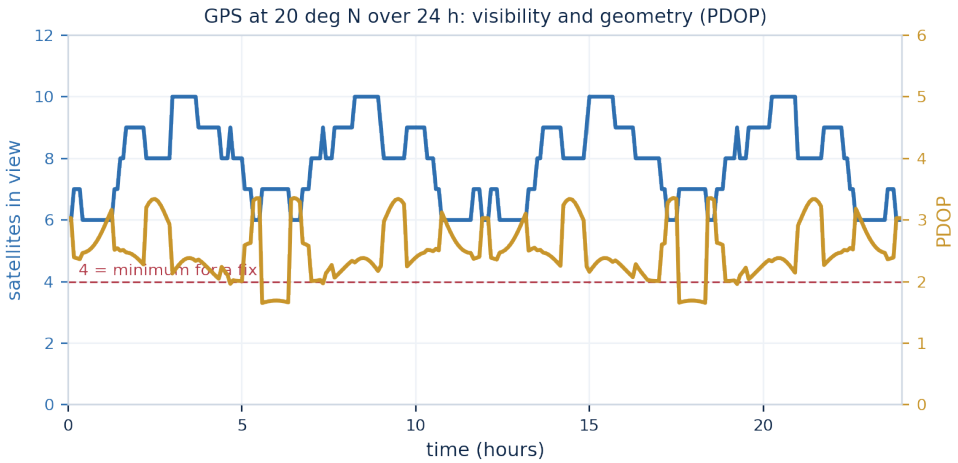


Figure 2. GPS at 20 degrees North over 24 hours. Six to ten satellites stay in view (blue), always above the four-satellite minimum, and the PDOP (gold) stays between about 1.6 and 3.4, so the geometry is consistently good. The two curves are the heartbeat of a navigation constellation.

PART 4

Constellation Dynamics and the Error Budget

Navigation needs the satellite positions known to better than a metre, so the orbital dynamics of earlier modules return with high stakes. The same perturbations, Earth oblateness (J2), luni-solar gravity and solar radiation pressure, must be modelled precisely and the orbits determined continuously from ground stations. The pieces that limit accuracy make up the error budget:

- **Ephemeris error:** imperfect knowledge of where the satellite actually is.
- **Satellite clock error:** residual after the broadcast clock correction.
- **Ionospheric and tropospheric delay:** the signal slows in the atmosphere; the largest correctable errors.
- **Multipath and receiver noise:** reflections and thermal noise at the user end.

Together these give the user-equivalent range error (UERE) that multiplies with DOP for the final position error. Augmentation systems drive it down: differential GNSS and SBAS broadcast corrections, while carrier-phase RTK reaches centimetre accuracy by resolving the signal wavelength itself.

PART 5

Relativity Is Not Optional

GNSS is the clearest everyday proof of relativity, because it fails without it. Two effects shift the satellite clocks relative to the ground. General relativity: the satellite sits higher in Earth gravity well, so its clock runs faster. Special relativity: the satellite moves fast, so its clock runs slower. The net is the difference:

$$\frac{\Delta t}{t} = \frac{\mu/r_g - \mu/r_s}{c^2}\ \text{(GR)}\ -\ \frac{v_s^2 - v_g^2}{2c^2}\ \text{(SR)}$$

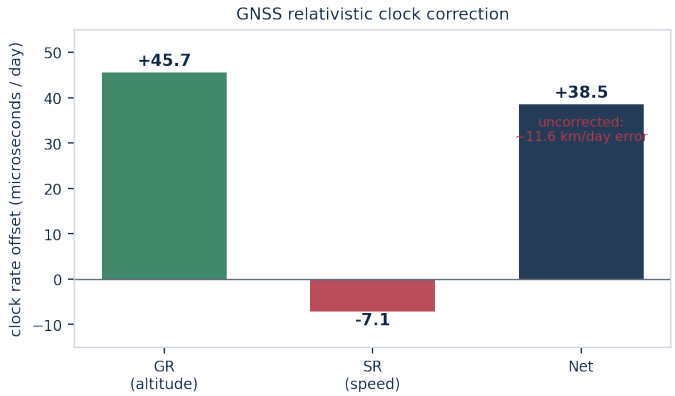


Figure 3. The two relativistic effects for GPS. The gravitational term (+45.7 microseconds per day) dominates the velocity term (-7.1), leaving a net of +38.5 microseconds per day.

Worked example: what relativity is worth

For GPS the net clock rate is +38.5 microseconds per day. Left uncorrected, that is a ranging error of 38.5 microseconds times the speed of light, about 11.6 kilometres per day, growing without bound. GNSS satellites carry a deliberately offset clock frequency and an on-board correction so that, seen from the ground, they keep true time. Get relativity wrong and the system is useless within hours.

Computing Visibility and DOP

The DOP analysis is short: propagate the constellation, find the satellites above the elevation mask, build the geometry matrix from their line-of-sight directions, and invert. This is exactly what produced Figure 2.

```
import numpy as np
# sats: ECEF positions of visible satellites; u: receiver ECEF; up: local vertical
def pdop(sats, u, up, mask_deg=5):
    rows = []
    for s in sats:
        e = (s - u); e /= np.linalg.norm(e)          # line of sight
        if np.degrees(np.arcsin(e @ up)) > mask_deg:  # above elevation mask
            rows.append([-e[0], -e[1], -e[2], 1.0])   # geometry row (+clock)
    H = np.array(rows)
    if len(H) < 4: return None                        # no fix
    Q = np.linalg.inv(H.T @ H)
    return np.sqrt(Q[0,0] + Q[1,1] + Q[2,2])          # PDOP
```

Production tools

STK ships a full GPS constellation and has Navigation Accuracy and Coverage tools that compute DOP, satellites in view and position-accuracy maps directly, using precise force models and real almanacs. GMAT and Python reproduce the geometry. As always, the hand relation position error equals DOP times UERE is the check on whatever the tool reports.

PRACTICE

This Week (aim for 12 to 15 hours)

Bring your work and plots to the next live session. As always, show your reasoning, not just the final numbers.

Assignment: GNSS Constellation Dynamics and Navigation

- **Reading and notes (3 to 4 h).** Explain why four satellites are needed for a fix, and what dilution of precision means geometrically. One page.
- **Constellation dynamics (3 to 4 h).** Compute the orbital period of a Galileo satellite (a = 29,600 km) and its revolutions per sidereal day, and compare to GPS. Explain what the difference means for ground-track repeat.
- **Relativity (3 to 4 h).** Compute the net relativistic clock rate for a Galileo satellite and the daily range error if it were uncorrected. Show the GR and SR parts separately.
- **DOP (2 h).** If a receiver has PDOP = 2.3 and the UERE is 4 m, what is the expected position error? List two ways to reduce DOP.
- **STK (2 h).** Load or build a GPS constellation, place a ground receiver, and plot satellites in view and PDOP over 24 hours. Capture the result.

Answer key (for your self-check)

Quantity	Expected value
Galileo period (a = 29,600 km)	about 14.08 h; 1.70 revs per sidereal day (vs GPS 11.97 h, 2.00)
Galileo relativistic net	GR +47.1, SR -6.4, net about +40.8 us/day, ~12.2 km/day uncorrected
Position error, PDOP 2.3, UERE 4 m	2.3 x 4 = 9.2 m
Reduce DOP	more satellites in view, or a wider spread across the sky (lower elevation mask)
GPS satellites in view (20 N)	6 to 10, PDOP about 1.6 to 3.4 over a day

Coming up in Session 9

Re-entry dynamics and mission end-of-life planning: the physics of atmospheric re-entry, the heating and deceleration a returning body faces, and how missions are safely disposed of at the end of their lives.

REFERENCE

One-Page Formula Sheet

Pseudorange measurement	$\rho_i = \lvert \mathbf{r}^i - \mathbf{r}_u \rvert + c\,\delta t_u + \varepsilon_i$
Unknowns and satellites needed	4 unknowns: $(x, y, z, \delta t_u) \Rightarrow$ need ≥ 4 satellites
Position error from geometry	$\sigma_{\text{pos}} = \text{PDOP} \times \sigma_{\text{UERE}}$
Position dilution of precision	$\text{PDOP} = \sqrt{Q_{xx} + Q_{yy} + Q_{zz}}, \quad Q = (H^T H)^{-1}$
Relativistic clock rate	$\frac{\Delta t}{t} = \frac{\mu/r_g - \mu/r_s}{c^2}$ (GR) $- \frac{v_s^2 - v_g^2}{2c^2}$ (SR)
GNSS orbit (MEO)	$T_{\text{GNSS}} \approx \frac{1}{2}$ sidereal day (GPS), $\quad a \approx 26{,}560$ km

Key glossary

- **Pseudorange:** measured range including the receiver clock offset, shared across all satellites.
- **DOP:** dilution of precision, the geometric factor that turns ranging error into position error.
- **UERE:** user-equivalent range error, the combined per-satellite ranging error.
- **SBAS / RTK:** augmentation systems that broadcast corrections or use carrier phase for higher accuracy.
- **Relativistic correction:** the +38.5 us/day clock offset GNSS must apply to keep true time.

Going further

For depth, see Misra and Enge, Global Positioning System: Signals, Measurements, and Performance, and Kaplan and Hegarty, Understanding GPS/GNSS Principles and Applications, plus Vallado for the orbital side. These are real, widely used texts; please confirm editions yourself, as I cannot verify those details from here.